Course Code	EE5130
Course Title	Artificial Intelligence and Machine Learning
No. of Credits	3
Pre-Requisites	EE2070 Random Signal Analysis
Compulsory / Optional	Optional

Aim (s): To provide the students with the knowledge and skills on solving engineering problems with concepts in Artificial Intelligence (AI) and Machine Learning (ML)

Intended Learning Outcomes:

At the end of this course, students should be able to:

LO1: **explain** the fundamental concepts, issues and challenges of AIML

LO2: **formalize** a given task as a AIML problem.

LO3: **identify** suitable algorithms to tackle different AIML problems by analyzing weaknesses and strengths of multiple algorithms.

LO4: **design** and Develop models and apply AIML frameworks to solve practical problems.

Course contents	**Time Allocated / hours**			
	L	**T**	**P**	**A**
1. Fundamental concepts in AI and ML: Introduction to AI and ML, Evolution of AI and ML, Types of Learning: Supervised, Unsupervised and Reinforcement Learning, Static and Dynamic Learning, Big Data and associated challenges, Trends and applications of AI and ML	2			
2. Linear methods in ML Representation and Visualization of Feature spaces, Linear and Nonlinear mapping of feature spaces, Curse of dimensionality and dimensionality reduction, Regularized and Generalized Regression Models, k-Nearest Neighbour (kNN) Algorithm	2			
3. Supervised Learning & Artificial Neural Networks (ANN): The challenge of mapping nonlinear feature spaces, The concept of perceptron, Multi Layer Perceptrons (MLP), Supervised Learning, and Feed Forward ANN, Overfitting and regularization, Radial Basis Function (RBF) ANNs, Echo-State ANN	3			
4. Unspervised Learning: Clustering, k-Means Algorithm, Component Analysis, Self Organizing Maps (SOM), Auto Encoding Neural Networks	2			

5. Deep Learning: Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Practical implementation of Deep Learning	2			
6. Decision Trees Regression and Classification Tree, Decision metrics, Chi-Squared Automatic Interaction Detection (CHAID) Algorithm, Ensemble Decision Trees (EDT), Boosted EDT (BEDT): AdaBoost and Gradient Boosting, Random Forest Trees and Decision Jungles,	2			
7. Support Vector Machines (SVM) Theory of SVM, Extension of SVM for Multi class and non-linea problems, Separability and Margins, Non-Linearity and use of Kernels, Risk Minimization	2			
8. Probabilistic Models Discriminative Models: Maximum Likelihood Estimation (MLE) and Baysian Approach, Mixture Models, Generative Models: Mixture Models, Baysian Networks	3			
9. Dynamic Programming and Reinforcement Learning Fundamentals of Dynamic programming and classes of problems, Characteristics and Frame work of Reinforcement learning	2			
10. Evolutionary Algorithms Genetic Algorithms, Swarm Intelligence, Ant Colony Optimization, Simulated annealing	4			
11. Time Series Models Stationarity, Auto Regressive (AR), Moving Avrage (MA), Auto Regressive Moving Avrage (ARMA) and Auto Regressive Integrated Moving Avrage (ARIMA) Models, Hidden Markov Models (HMM), Conditional Random Fields (CRF)	3			
12. Emerging Trends in Machine Learning Transfer learning, Generative Adversarial Networks (GAN), Quantum Computation: Quantum theory, Entanglement and Superposition, Computation with quantum particles, Auto ML	3			
13. Implementation platforms for AI & ML				6
14. AIML Mini Project				24
Total Hours (Notional Hours 150)	30			30

Recommended Texts:

1. Simon Haykin. Neural Networks and Learning Machines, 3rd Edition, Prentice Hall, 2008
2. Christoper M. Bishops, Pattern Recognition and Machine Learning, Springer 2006
3. Trevor Hastie, Robert Tibshirani and Jerome Friedman, Elements of Statistical Learning, Springer 2009
4. Shai Shalev-Shwartz, Shai Ben-David, Understanding Machine Learning: From Theory to Algorithms, Cambridge University Press, 2014

Assessment	**Percentage Mark**
Continuous Assessment Tutorials Assignments Practical Work	 40 10
End Semester	50

Note: L – Lectures, T – Tutorials, P – Practical, A - Assignments